Record-Time Compactness Solves the Asymptotic Eigenvector Conjecture

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Abstract

Akian, Gaubert, and Nussbaum conjectured that the bounded-orbit hypothesis in their asymptotic eigenvector theorem for continuous homogeneous order-preserving maps on sup-semilattice cones is unnecessary. We prove the conjecture as stated. The key is to normalize a super-eigenvector orbit at carefully chosen record times of its running norm maximum. Backward shifts of the normalized record vectors remain in the unit ball. A strict contraction of a measure of noncompactness then makes the record vectors precompact. A limit is a super-eigenvector with bounded forward orbit, so the conditional part of the source theorem applies.

1 Source problem

Let C be a proper cone in a Banach space X, and let $h : C \to C$ be continuous, positively homogeneous, and order-preserving. Theorem 5.1 of Akian–Gaubert–Nussbaum [1] assumes that C is a sup-semilattice and that

$$\rho(h, C) < r(h, C),$$

where $r(h, C)$ is the cone spectral radius and $\rho(h, C)$ is the cone essential spectral radius associated with a fixed homogeneous generalized measure of noncompactness. The theorem first produces $z \in C \setminus \{0\}$ with $h(z) \geq r(h, C)z$. It obtains an eigenvector at $r(h, C)$ if, additionally, some such super-eigenvector has bounded normalized forward orbit. Remark 5.2 conjectures that this extra condition is unnecessary.

The next source crop contains Theorem 5.1 (PDF page 10). Its continuation and Remark 5.2 appear on the following page.

since $\|h^n\|_C < 1$, we deduce that $x = 0$. $\square$

5. Existence of eigenvectors of order-preserving maps in sup-semilattices

Let C be a proper cone in a Banach space X. We say that a subset D of X is a *sup-semilattice* (in the ordering from C) if, for all $x, y \in D$, there exists $z \in D$ such that $x \leqslant z, y \leqslant z$ and $z \leqslant w$ for every $w \in D$ for which $x \leqslant w$ and $y \leqslant w$. Then, the element z as above is unique and we shall write it $w = x \vee y$. If C is not normal, it may easily happen (see the remark after Lemma 3.6 in [MPN02]) that X is a sup-semilattice but that the map $(x, y) \mapsto x \vee y$ is not continuous. The notion of *inf-semilattice* is defined dually (by reversing the order). A *lattice* is an ordered set which is both an inf-semilattice and a sup-semilattice.

Theorem 5.1. *Let C be a proper cone in a Banach space $(X, \|\cdot\|)$, and assume that C is a sup-semilattice in the partial ordering induced by C. Assume that $h : C \to C$ is continuous, homogeneous and order-preserving in the partial ordering from C, and that $\rho(h\,, C) < r(h\,, C)$. Then there exists $z \in C \setminus \{0\}$ such that*

$$h(z) \geqslant r(h\,, C)z \ .$$

If, in addition, we assume that there exists $\zeta \in C \setminus \{0\}$ such that

$$\text{(a)} \qquad h(\zeta) \geqslant r(h\,, C)\zeta, \qquad \textit{and}$$

$$\text{(b)} \qquad \{\frac{h^k(\zeta)}{r(h\,, C)^k} \mid k \geqslant 0\} \textit{ is bounded,}$$

then $h(x) = r(h\,, C)x$ for some $x \in C \setminus \{0\}$, and, in particular, $\hat{r}(h\,, C) = r(h\,, C)$.

Proof. By replacing h by $r^{-1}h$, where $r = r(h\,, C)$, we can assume that $r = r(h\,, C) = 1$. Since $\rho(h\,, C) < r(h\,, C) = 1$, there exists $N \geqslant 1$ such that $\nu(h^m\,, C) < 1$, for all $m \geqslant N$. Select a fixed $m \geqslant N$. Theorem 3.4 implies that there exists $x_m \in C \setminus \{0\}$ such that $h^m(x_m) = x_m$. Define $z := x_m \vee h(x_m) \vee \cdots \vee h^{m-1}(x_m)$, so $z \geqslant h^j(x_m)$ for all $j \geqslant 0$. It follows that $h(z) \geqslant h^{j+1}(x_m)$ for all $j \geqslant 0$ (h is order-preserving), and since $h^m(x_m) = x_m$, $h(z) \geqslant h^j(x_m)$ for all $j \geqslant 0$. By the properties of the sup-semilattice, we must have $h(z) \geqslant z$.

Now assume that there exists $\zeta \in C \setminus \{0\}$ satisfying (a) and (b). This implies that the sequence $(h^k(\zeta))_{k \geqslant 0}$ is nondecreasing. Let $S := \{h^k(\zeta) \mid k \geqslant 0\}$, and for $0 \leqslant j \leqslant m-1$, let $S_j := \{h^{km+j}(\zeta) \mid k \geqslant 0\}$. Notice that we have $S_j = \{h^j(\zeta)\} \cup h^m(S_j)$. Since S_j is assumed to be bounded, we obtain, using (3.5), that

$$\nu(S_j) = \nu(h^m(S_j)) \leqslant \nu(h^m\,, C)\nu(S_j) \ ;$$

and since $\nu(h^m\,, C) < 1$, we conclude that $\nu(S_j) = 0$, and $\operatorname{cl} S_j$ is compact. It follows that $S = \bigcup_{j=0}^{m-1} S_j$ has compact closure. Because $\operatorname{cl} S$ is compact, there exists a strictly increasing sequence of integers k_i with $h^{k_i}(\zeta) \to w$. We claim that $h^k(\zeta)$ converges towards w as $k \to \infty$. If not, there exists a sequence $\ell_i \to \infty$ with $\|h^{\ell_i}(\zeta) - w\| \geqslant \delta > 0$; and by taking a further subsequence we can assume that $h^{\ell_i}(\zeta) \to w'$, $w' \neq w$, as $i \to \infty$. For any fixed i, we have $k_j \geqslant \ell_i$ for sufficiently large

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j, so $h^{k_j}(\zeta) \geqslant h^{\ell_i}(\zeta)$ for all j large and $w \geqslant h^{\ell_i}(\zeta)$. If we now let i approach infinity we see that $w \geqslant w'$. By symmetry of this argument, we also obtain $w' \geqslant w$; so $w = w'$. This contradicts our original assumption, so we must have that $h^k(\zeta) \to w$ as $k \to \infty$ and $h^{k+1}(\zeta) \to w$ as $k \to \infty$. However, the continuity of h at w implies that $h^{k+1}(\zeta) \to h(w)$, so $h(w) = w$. Note that $w \geqslant \zeta$, so $w \neq 0$. □

Remark 5.2. We conjecture that if hypotheses are as above, but we do not assume the existence of $\zeta \in C \setminus \{0\}$ satisfying (a) and (b) in Theorem 5.1, it is still true that $h(x) = r(h, C)x$ for some $x \in C \setminus \{0\}$. In general, fixed point theorems which deduce the existence of fixed points of a given map f by making assumptions about the behavior of (large) iterates f^m of f have been called "asymptotic fixed point theorems"; see [Nus72, Nus77, Nus85] and references to the literature there. Proving asymptotic fixed point theorems is sometimes surprisingly difficult, and a number of old fundamental conjectures (see [Nus72, Nus77, Nus85]) remain open. Our conjecture here fits into this framework, inasmuch as the assumption $\rho(h, C) < r(h, C) = 1$ is an assertion about the behavior of iterates of h and not directly about h.

The hypotheses of Theorem 5.1 may seem difficult to verify. However, we shall

Transcription of Remark 5.2. Under the preceding hypotheses, but without the extra vector satisfying conditions (a) and (b) of Theorem 5.1, the authors conjecture that there is still $x \in C \setminus \{0\}$ such that $h(x) = r(h, C)x$ [1, Remark 5.2].

2 Definitions and proof intuition

For a positively homogeneous self-map h of C, the source defines

$$r(h, C) = \sup_{u \in C} \limsup_{n \to \infty} \|h^n(u)\|^{1/n}.$$

Let ν be the homogeneous generalized measure of noncompactness used in the source. We use three of its stated properties:

1. $\nu(A) = 0$ if and only if $\operatorname{cl} A$ is compact;
2. $A \subseteq B$ implies $\nu(A) \leq \nu(B)$;
3. adjoining or deleting a relatively compact set, in particular finitely many points, does not change ν [1, Proposition 3.3].

The coefficient $\nu(g, C)$ is defined so that

$$\nu(g(A)) \leq \nu(g, C)\nu(A)$$

for every bounded $A \subseteq C$. The definition of the essential spectral radius implies that $\rho(h, C) < 1$ yields an $m \geq 1$ with $\nu(h^m, C) < 1$.

Proof intuition. The source already supplies a super-eigenvector $z \le h(z)$. Its orbit can be unbounded, but at spectral radius one it grows only subexponentially. Take the running maximum $A_n = \max_{j\le n} \|h^j(z)\|$. Subexponential growth forces arbitrarily long almost-flat windows of A_n. Choosing the left endpoint at a genuine orbit-norm record gives unit vectors x_i.

To recover compactness without assuming a bounded original orbit, look backward from each record time. Every fixed backward shift remains in the unit ball because the denominator is the later record maximum. Applying a strictly condensing iterate h^m shifts one backward-record set onto the preceding one, modulo a single point. Iterating the measure-of-noncompactness inequality from arbitrarily far backward forces the record-vector set to have measure zero. The almost-flat forward window then shows that a limit of the record vectors has a bounded forward orbit.

3 Two lemmas

Lemma 1 (Flat record times)**.** *Let $(A_n)_{n\ge 0}$ be a positive, nondecreasing, unbounded sequence with*

$$\lim_{n\to\infty} A_n^{1/n} = 1.$$

Suppose $A_n = \max_{0\le j\le n} a_j$ for a positive sequence (a_j). For every fixed positive integer m, there are strictly increasing integers $k_i > im$ such that

$$a_{k_i} = A_{k_i}, \qquad 1 \le \frac{A_{k_i+i}}{A_{k_i}} < e^{1/i}. \tag{1}$$

Proof. For every positive integer L and every $\eta > 0$, there are arbitrarily large n such that

$$A_{n+L} < e^{\eta} A_n. \tag{2}$$

Indeed, if the reverse inequality held for all sufficiently large n, iteration along one arithmetic progression of step L would give

$$\liminf_{n\to\infty} A_n^{1/n} \ge e^{\eta/L} > 1,$$

a contradiction.

Inductively take n_i satisfying (2) with $L = i$ and $\eta = 1/i$, so large that A_{n_i} exceeds every a_j with $j \le \max\{im, k_{i-1}\}$. Choose $k_i \le n_i$ with $a_{k_i} = A_{n_i}$. Then $k_i > \max\{im, k_{i-1}\}$, and $A_{k_i} = A_{n_i}$. If $k_i + i \le n_i$, monotonicity gives $A_{k_i+i} = A_{k_i}$. Otherwise $k_i + i \le n_i + i$, so

$$\frac{A_{k_i+i}}{A_{k_i}} \le \frac{A_{n_i+i}}{A_{n_i}} < e^{1/i}.$$

This proves (1). $\square$

Lemma 2 (Compactness from backward records)**.** *Let $z_n = h^n(z)$, let $A_n = \max_{0\le j\le n} \|z_j\|$, and suppose the indices k_i satisfy Lemma 1 with $a_j = \|z_j\|$. If $c := \nu(h^m, C) < 1$, then*

$$\left\{ \frac{z_{k_i}}{A_{k_i}} : i \ge 1 \right\}$$

has compact closure.

Proof. For every integer $q \geq 0$, set

$$E_q = \left\{ \frac{z_{k_i - qm}}{A_{k_i}} : i \geq q+1 \right\}.$$

The indices are nonnegative because $k_i > im$. Also E_q is contained in the closed unit ball B_X: for $i \geq q+1$,

$$\left\| \frac{z_{k_i - qm}}{A_{k_i}} \right\| \leq \frac{A_{k_i}}{A_{k_i}} = 1.$$

Positive homogeneity gives

$$E_q = h^m(E_{q+1}) \cup \left\{ \frac{z_{k_{q+1} - qm}}{A_{k_{q+1}}} \right\}.$$

Finite-set invariance of ν and the definition of c imply

$$\nu(E_q) = \nu(h^m(E_{q+1})) \leq c\nu(E_{q+1}).$$

Since every $E_q \subseteq B_X$, iteration yields, for every $N \geq 1$,

$$\nu(E_0) \leq c^N \nu(E_N) \leq c^N \nu(B_X).$$

The number $\nu(B_X)$ is finite and $c < 1$, hence $\nu(E_0) = 0$. The defining compactness property of ν completes the proof. □

4 Full solution

Theorem 1. *Let C be a proper cone in a Banach space X, and assume that C is a sup-semilattice in its induced order. Let $h : C \to C$ be continuous, positively homogeneous, and order-preserving. If*

$$\rho(h, C) < r(h, C),$$

then there is $y \in C \setminus \{0\}$ such that

$$h(y) = r(h, C)y.$$

Thus the conjecture in Remark 5.2 of [1] holds in full.

Proof. Put $r = r(h, C)$. The strict spectral gap implies $r > 0$. Replacing h by $r^{-1}h$, we may assume

$$r(h, C) = 1, \qquad \rho(h, C) < 1. \tag{3}$$

The first conclusion of Theorem 5.1 in [1] gives $z \in C \setminus \{0\}$ such that $z \leq h(z)$. Let

$$z_n = h^n(z), \qquad A_n = \max_{0 \leq j \leq n} \|z_j\|.$$

The sequence (z_n) is increasing in the cone order. By the definition of the cone spectral radius and (3),

$$\limsup_{n \to \infty} \|z_n\|^{1/n} \leq 1.$$

It follows that

$$\lim_{n\to\infty} A_n^{1/n} = 1. \tag{4}$$

For completeness, the upper bound follows because, for every $\varepsilon > 0$, all sufficiently late z_j satisfy $\|z_j\| \le (1+\varepsilon)^j$, while the finitely many earlier norms are absorbed by a constant. The lower bound follows from $A_n \ge \|z\| > 0$.

If (A_n) is bounded, then the orbit $(h^n(z))$ is bounded. Conditions (a) and (b) of Theorem 5.1 are satisfied by z, and that theorem gives a nonzero fixed point of h. Assume from now on that (A_n) is unbounded.

By (3), choose $m \ge 1$ such that

$$c := \nu(h^m, C) < 1. \tag{5}$$

Apply Lemma 1 to $a_j = \|z_j\|$, and set

$$x_i = \frac{z_{k_i}}{A_{k_i}}.$$

Since $\|z_{k_i}\| = A_{k_i}$, every x_i has norm one. Lemma 2 shows that (x_i) has a norm-convergent subsequence. Relabel it so that $x_i \to x$. Then $x \in C$ and $\|x\| = 1$.

Since $z_{k_i} \le z_{k_i+1}$, we have $x_i \le h(x_i)$. The cone is closed and h is continuous, so

$$x \le h(x). \tag{6}$$

It remains to show that the forward orbit of x is bounded. Fix $j \ge 0$. For all sufficiently large $i \ge j$, (1) gives

$$\|h^j(x_i)\| = \frac{\|z_{k_i+j}\|}{A_{k_i}} \le \frac{A_{k_i+i}}{A_{k_i}} < e^{1/i}.$$

Continuity of the fixed iterate h^j and $x_i \to x$ therefore imply

$$\|h^j(x)\| \le 1 \qquad (j \ge 0). \tag{7}$$

Thus $x \ne 0$, (6) is the super-eigenvector condition, and (7) is the bounded-orbit condition in the second part of Theorem 5.1. That theorem now provides $y \in C \setminus \{0\}$ with $h(y) = y$. Undoing the normalization gives $h(y) = r(h, C)y$. □

5 Verification, scope, and novelty boundary

The proof uses neither normality of the cone nor norm monotonicity. The running maximum A_n, rather than $\|z_n\|$ itself, is what makes all backward record shifts uniformly bounded. It also uses no uniform continuity on bounded sets: only continuity of each fixed iterate is needed after the compact subsequence is obtained.

The local result ledger, solution and attempt indexes, proof-gap index, and full parsed arXiv source corpus were searched for arXiv:1112.5968 and the phrases "Remark 5.2", "asymptotic fixed point", "bounded orbit", "record time", and "cone essential spectral radius". An external exact-phrase search and the 30-work OpenAlex citation set for the source were also checked, including the nearby works arXiv:1302.3905, arXiv:1406.6657, arXiv:1612.01755, and the 2017 paper of Thieme on order-bounded maps. No later statement resolving Remark 5.2, and no matching record-time proof, was found. This was a bounded search rather than a systematic review, so novelty remains subject to expert literature review.

6 Human review focus

Review is recommended in the following order:

1. verify the arbitrarily late flat-window assertion (2);
2. verify that choosing an actual record index preserves the flat window;
3. check the identity expressing E_q as $h^m(E_{q+1})$ with one indexed point adjoined;
4. check that all E_q lie in the same unit ball, uniformly in q;
5. verify the fixed-j limiting argument that yields the bounded orbit in (7).

References

[1] M. Akian, S. Gaubert, and R. Nussbaum, *A Collatz–Wielandt characterization of the spectral radius of order-preserving homogeneous maps on cones*, arXiv:1112.5968. https://arxiv.org/abs/1112.5968